Retrieval Evaluation Results

Chunk / Overlap	Device	Precision@3	Recall@3	MRR@3
800 / 80	CPU	0.4722	0.7917	0.5556
800 / 80	GPU	0.4722	0.7917	0.5556
500 / 50	CPU	0.3611	0.6528	0.6528
500 / 50	GPU	0.3611	0.6528	0.6528
1500 / 300	CPU	0.5833	0.75	0.7778
1500 / 300	GPU	0.5833	0.75	0.7778
300 / 60	CPU	0.4167	0.6389	0.6111
300 / 60	GPU	0.4167	0.6389	0.6111

Comparisons and Justification

Chunk / Overlap	Device	Precision@3	Recall@3	MRR@3
800 / 80	CPU	0.4722	0.7917	0.5556
800 / 80	GPU	0.4722	0.7917	0.5556
VS				
Chunk / Overlap	**Device**	**Precision@3**	**Recall@3**	**MRR@3**
1500 / 300	CPU	0.5833	0.75	0.7778
1500 / 300	GPU	0.5833	0.75	0.7778

Although the 1500/300 configuration achieved higher Precision\@3, Recall\@3, and MRR\@3 scores, the 800/80 configuration was selected because the evaluation also considered the actual relevance and usefulness of the retrieved passages, not just the numerical results. During the assessment, 800/80 was able to retrieve more suitable information for certain questions, whereas 1500/300 did not consistently provide the most relevant answers despite its higher scores. Therefore, 800/80 was considered a more appropriate balance between retrieval performance and answer relevance for the project's requirements.

CPU AND GPU PERFORMANCE RESULTS

Device	Chunk Size	Overlap	Top-K	Indexing Time	Average Response Time	Total Execution Time	Results
CPU	800	80	3	1 min 5.57 sec	32.75 sec	19 min 26.56 sec	GPU is faster.
GPU	800	80	3	1 min 1.29 sec	36.18 sec	9 min 48.53 sec	GPU is faster.
CPU	500	50	3	1 min 11.19 sec	24.34 sec	10 min 21.06 sec	Almost Identical
GPU	500	50	3	1 min 18.60 sec	25.41 sec	10 min 21.50 sec	Almost Identical
CPU	1500	300	3	1 min 0.13 sec	44.00 sec	11 min 5.25 sec	GPU is faster.
GPU	1500	300	3	5.48 sec	2.71 sec	2 min 53.79 sec	GPU is faster.
CPU	300	60	3	1 min 33.83 sec	23.74 sec	8 min 10.49 sec	GPU is faster.
GPU	300	60	3	2.62 sec	2.18 sec	2 min 46.30 sec	GPU is faster.

Selected Configuration

Parameter	Value
Chunk size	800 characters
Chunk overlap	80 characters
Top-K	3 document passages
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector store	FAISS index
Hardware accelerator	Google Colab NVIDIA T4 GPU
Precision@3/Recall@3/MRR@3	0.4722/0.7917/0.5556

Results for all 12 questions

The values in each cell follow this order:

Precision@3 / Recall@3 / MRR@3

Question	800/80 CPU	800/80 GPU	500/50 CPU	500/50 GPU	1500/300 CPU	1500/300 GPU	300/60 CPU	300/60 GPU
Q1	.67 / 1 / .50	.67 / 1 / .50	.67 / 1 / 1	.67 / 1 / 1	1 / 1 / 1	1 / 1 / 1	.33 / 1 / .33	.33 / 1 / .33
Q2	.33 / 1 / .33	.33 / 1 / .33	.33 / .67 / 1	.33 / .67 / 1	.33 / 1 / 1	.33 / 1 / 1	.67 / 1 / 1	.67 / 1 / 1
Q3	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1	.33 / 1 / .50	.33 / 1 / .50
Q4	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
Q5	.67 / .50 / .50	.67 / .50 / .50	0 / 0 / 0	0 / 0 / 0	1 / 1 / 1	1 / 1 / 1	0 / 0 / 0	0 / 0 / 0
Q6	.33 / 1 / .33	.33 / 1 / .33	.33 / 1 / .50	.33 / 1 / .50	.33 / .50 / .33	.33 / .50 / .33	.33 / .50 / 1	.33 / .50 / 1
Q7	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
Q8	.33 / 1 / .50	.33 / 1 / .50	.33 / 1 / 1	.33 / 1 / 1	.33 / 1 / 1	.33 / 1 / 1	1 / 1 / 1	1 / 1 / 1
Q9	.67 / 1 / 1	.67 / 1 / 1	.33 / 1 / 1	.33 / 1 / 1	.67 / 1 / 1	.67 / 1 / 1	.33 / .67 / 1	.33 / .67 / 1
Q10	.33 / 1 / .50	.33 / 1 / .50	.67 / 1 / 1	.67 / 1 / 1	.67 / 1 / 1	.67 / 1 / 1	.67 / 1 / 1	.67 / 1 / 1
Q11	.67 / 1 / 1	.67 / 1 / 1	.33 / .67 / 1	.33 / .67 / 1	1 / 1 / 1	1 / 1 / 1	.67 / 1 / .50	.67 / 1 / .50
Q12	.67 / 1 / 1	.67 / 1 / 1	.33 / .50 / .33	.33 / .50 / .33	.67 / .50 / 1	.67 / .50 / 1	.67 / .50 / 1	.67 / .50 / 1